AgriDirect AI

Crop demand prediction · Direct farmer-to-company marketplace

RESEARCH REPORT

Market, Competitor & Data Feasibility Research

What exists today, where AgriDirect AI fits, and whether the data is usable

Version	**1.0**
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Research window	**Public sources up to October 2026**
Audience	**Project lead, developers, college reviewers**
Status	**Living document — update after user interviews**

Confidential — prepared for the AgriDirect AI development team.

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1. Executive summary

- **The problem is real and recognised.** The government's own e-NAM platform has grown to 1,656 mandis and about 1.80 crore registered farmers (March 2026), showing demand for digital market access — but e-NAM still needs produce to be physically brought to a registered mandi.
- **Large agritech players focus elsewhere.** DeHaat and AgroStar are input- and advisory-led; Ninjacart and WayCool run their own procurement supply chains. None offers an open, verified *discovery* layer where a farmer lists before harvest and nearby companies post requirements, combined with pre-sowing demand prediction.
- **Data is available but fragile.** Agmarknet price/arrival data is published under the Open Government License (India) and covers 300+ commodities across thousands of markets, but the official API is unreliable; a historical mirror (CEDA, Ashoka University) and admin CSV upload are needed as fallbacks.
- **Weather data is free only for non-commercial use.** Open-Meteo suits the prototype and pilot research; a commercial launch needs its paid plan or NASA POWER / IMD.
- **Privacy law is now concrete.** DPDP Rules 2025 were notified on 13 November 2025; most operational duties apply from mid-May 2027 — before AgriDirect AI would reach scale, so build compliance in from day one.
- **Recommendation:** pilot in one Pune-division district with 3–5 data-rich crops, seed demand with FPOs and local processors, and lead the pitch with the matching marketplace (clear value) while presenting predictions as advisory.

2. Market context

2.1 Government digital markets

e-NAM, launched April 2016 and implemented by SFAC, integrates physical APMC mandis into an online trading platform. By March 2026 it connected 1,656 mandis across 23 states and 4 UTs, with about 1.80 crore farmers, 2.73 lakh traders and 4,724 FPOs registered, and cumulative trade of roughly ₹4.8 lakh crore.

> **Implication:** farmers and FPOs are already used to registering on digital trade platforms. But e-NAM bidding happens on lots that are physically brought, weighed and graded at a registered mandi. AgriDirect AI can sit *before* the mandi — listing expected produce weeks ahead and connecting it to buyers who plan procurement.

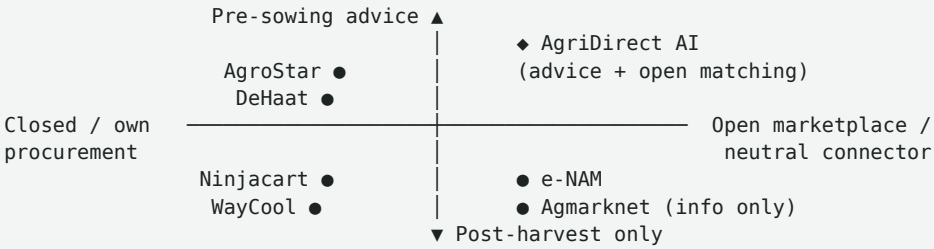
2.2 Pain points confirmed by sources

- Information asymmetry: small farmers depend on traders and commission agents for price information.
- Coordination failure: collective planting of the same high-price crop causes harvest gluts (recurring in onion, tomato).
- Remote-trade barriers: weak logistics for inter-state delivery, low digital literacy, dispute-resolution gaps, and cash/intermediary-dominated practices persist in many mandis.
- Buyers: procurement teams lack visibility of nearby supply and quality before delivery.

3. Competitor analysis

Player	What they do	Strength	Gap AgriDirect AI targets
e-NAM (Govt.)	Online bidding across integrated APMC mandis	Scale, trust, direct payment to farmer bank accounts	No pre-harvest listing or crop-choice advice; produce must reach the mandi
DeHaat	Full-stack: inputs, advisory, market linkage for small farmers in ~12 states	Largest agritech by revenue (₹3,000 cr+ FY25); physical centres	Closed ecosystem; strongest in east/north India; not an open buyer marketplace
AgroStar (Pune)	Agri-input e-commerce + advisory app with vernacular support	5M+ farmer network incl. Maharashtra; raised ₹725 cr in Jan 2026	Input-sales model; no buyer-requirement matching
Ninjacart	B2B fresh-produce supply chain: buys from farmers, sells to retailers	Logistics, buyer credit, large funding	Metro/tier-1 focus; platform is the buyer, not a neutral connector
WayCool	Full-stack agri supply chain connecting farmers to retailers/processors	Processing and distribution network	Own procurement; limited Maharashtra discovery for independent buyers
Vegrow	Advisory, grading, packaging, logistics and sales support for fruit growers	Quality grading	Fruit-focused; not a requirement-matching marketplace
Agmarknet / Kisan apps	Price information only	Free, official	No recommendation, no buyer connection

3.1 Positioning map



3.2 Differentiators to defend

1. **Before-harvest listing** — produce is discoverable weeks ahead, so buyers can plan.
2. **Requirement-first matching** — buyers state need once; system pushes suitable farmers.
3. **Demand-aware crop advice** — uses live platform requirements as a demand signal no public dataset has.
4. **Neutral connector** — the platform does not buy or resell, keeping trust with both sides.
5. **Marathi-first UX** for the Maharashtra pilot.

4. Data source feasibility

Source	What it gives	Access & licence	Risk	Verdict
Agmarknet via data.gov.in	Daily min/max/modal price + arrivals; 300+ commodities, ~4,500 markets (Jan 2025)	Free API key; Open Government License – India	API availability inconsistent; some resources show 'API does not exist'	Use, with fallback
CEDA Agri Market (Ashoka Univ.)	Agmarknet history from 2000, district aggregates	Free, research-oriented	Lags by a few months; check terms for commercial use	Use for training history
Open-Meteo	16-day forecast, history from 1940 (ERA5)	Free non-commercial (~10k calls/day); data CC BY 4.0; paid plan for commercial	Licence change needed at launch	Use for prototype/pilot
NASA POWER	Daily agro-climatology (rain, temp, solar)	Free, public	Coarse resolution (~0.5°)	Backup / commercial-safe
IMD	Official forecasts and district advisories	Mixed; some APIs need agreement	Access process slow	Phase 2
ISRIC SoilGrids	250 m soil properties (pH, carbon, texture)	CC BY 4.0	Modelled, not field-measured	Use
Soil Health Card	Field soil test summaries	Public portal, aggregated	Format inconsistent	District summaries only
ICAR / state crop calendars	Sowing/harvest windows, crop suitability	Public documents	Manual curation	Encode as rules

Phase 0 action: in week 1, pull 3 years of maize, soybean and onion prices for the pilot district from both data.gov.in and CEDA, compare coverage gaps, and record the result in the feasibility note. If both fail, the admin CSV upload becomes a Must-have.

5. Technical research notes

5.1 Forecasting approaches for mandi prices

Approach	Pros	Cons	Use in MVP
Seasonal naive	Trivial, strong baseline for seasonal crops	No external signals	Mandatory baseline
SARIMAX	Interpretable, handles seasonality + exogenous rain	One model per series, slow to scale	Comparison
Prophet	Easy holidays/seasonality	Often beaten by GBMs on noisy daily prices	Comparison
LightGBM / XGBoost on lag features	One global model across crops and markets; fast; SHAP explanations	Needs careful leakage control	Primary
LSTM / Temporal Fusion Transformer	Can capture complex patterns	Data-hungry, harder to explain, heavier to serve	Phase 2 research

5.2 Key lessons

- Forecast **direction and range**, not an exact price — easier to get right and easier for farmers to act on.
- Arrival volumes are often a leading indicator of price falls; include arrival anomalies as features.
- Validate with walk-forward splits by season; random splits overstate accuracy.
- Show uncertainty plainly (Low / Medium / High confidence) to avoid over-trust.

6. Pilot recommendation

Decision	Recommendation	Why
Region	One district in Pune division (e.g., Pune or Ahilyanagar)	Dense APMC network, processors and FPOs nearby, team is Pune-based
Crops (3–5)	Maize, soybean, onion, tur (pigeon pea), tomato	Good Agmarknet coverage; clear industrial buyers (feed, oil, processing); known price volatility
Buyers to seed	2–3 feed/oil processors, 1–2 FPOs, 1 retailer/aggregator	Solves cold start; real requirements feed the demand signal
Farmers to seed	30–50 via FPO and college network	Enough for meaningful matches
Success test	10 accepted deals in 8 weeks	Proves matching value independent of model accuracy

7. User research plan

Before freezing the UI, interview 8 farmers and 4 buyers using the clickable prototype.

7.1 Farmer interview guide

1. How did you decide what to plant this season? Who did you ask?
2. Last harvest: whom did you sell to, how was the price decided, how long did payment take?
3. Show crop-advice screen: what do you understand from it? Would you trust it? What would make you trust it more?
4. Show create-listing screen: complete it yourself (observe where you hesitate).
5. Would you prefer SMS, WhatsApp or app notifications? Which language?

7.2 Buyer interview guide

1. How do you source this crop today? How many intermediaries?
2. What quality parameters matter (moisture, grade, size)?
3. Show post-requirement and browse screens: what is missing for you to send an offer?
4. Would you pay a subscription or per-deal fee for verified nearby supply?

7.3 Usability metrics

- Task success: farmer creates a listing unaided (target 80%).
- Time on task: listing in under 2 minutes.
- Comprehension: farmer correctly explains the advice card in their own words (target 70%).

8. Business model options

Model	How it works	Pros	Cons
Free for farmers, subscription for companies	₹1,500–5,000/month per buyer for posting + search	Aligns with who gets procurement value; farmers onboard freely	Needs enough supply to justify
Commission per deal	0.5–1% of deal value from buyer	Revenue scales with usage	Hard to enforce since payment is off-platform
Freemium + premium insights	Advanced forecasts and alerts for FPOs/traders	Monetises the AI	Must prove accuracy first
Grants / CSR / FPO partnerships	Agri-innovation grants, state schemes	Fits pilot stage	Not sustainable alone

Suggested path: free during pilot → company subscription after 10+ successful deals per district.

9. SWOT

Strengths	Weaknesses
Combines advice + matching; neutral connector; Marathi-first; low-cost stack	Depends on fragile public data; cold start; no logistics/payments in MVP

Opportunities	Threats
FPO growth; processor demand for traceable supply; e-NAM familiarity	Well-funded incumbents adding similar features; deals moving off-platform; data licence changes

10. Sources

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> Figures are as reported by the cited sources on the dates shown; verify before using in investor or client material.